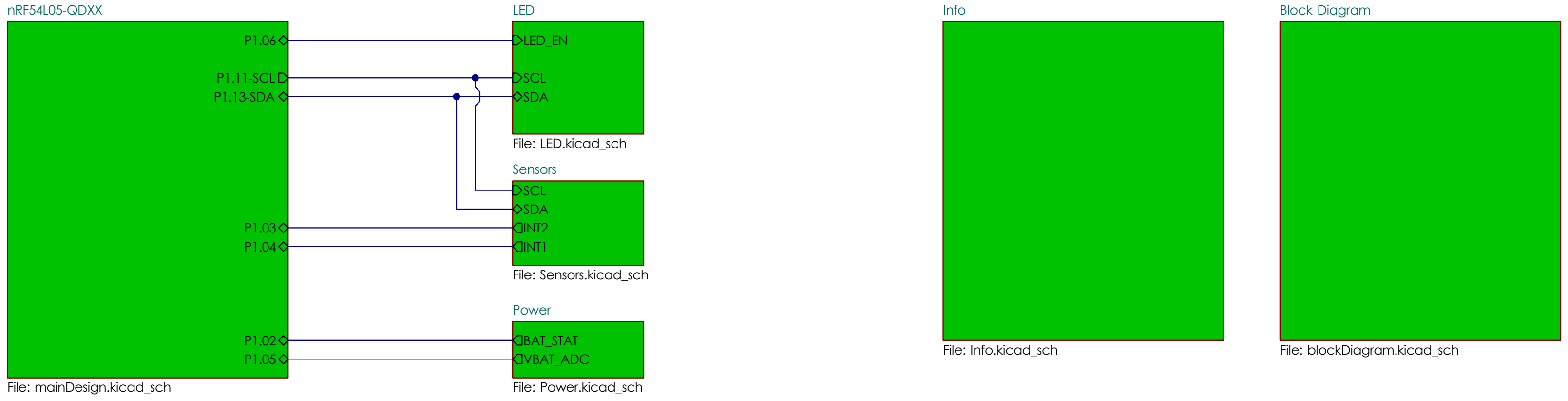

nRF54L05-QDXX
P1.06
P1.11-SCL
P1.13-SDA
P1.03
P1.04
P1.02
P1.05
File: mainDesign.kicad_sch
LED
LED_EN
SCL
SDA
File: LED.kicad_sch
Sensors
SCL
SDA
INT2
INT1
File: Sensors.kicad_sch
Power
BAT_STAT
VBAT_ADC
File: Power.kicad_sch
Info
File: Info.kicad_sch
Block Diagram
File: blockDiagram.kicad_sch

Sheet: /
File: Board.kicad_sch

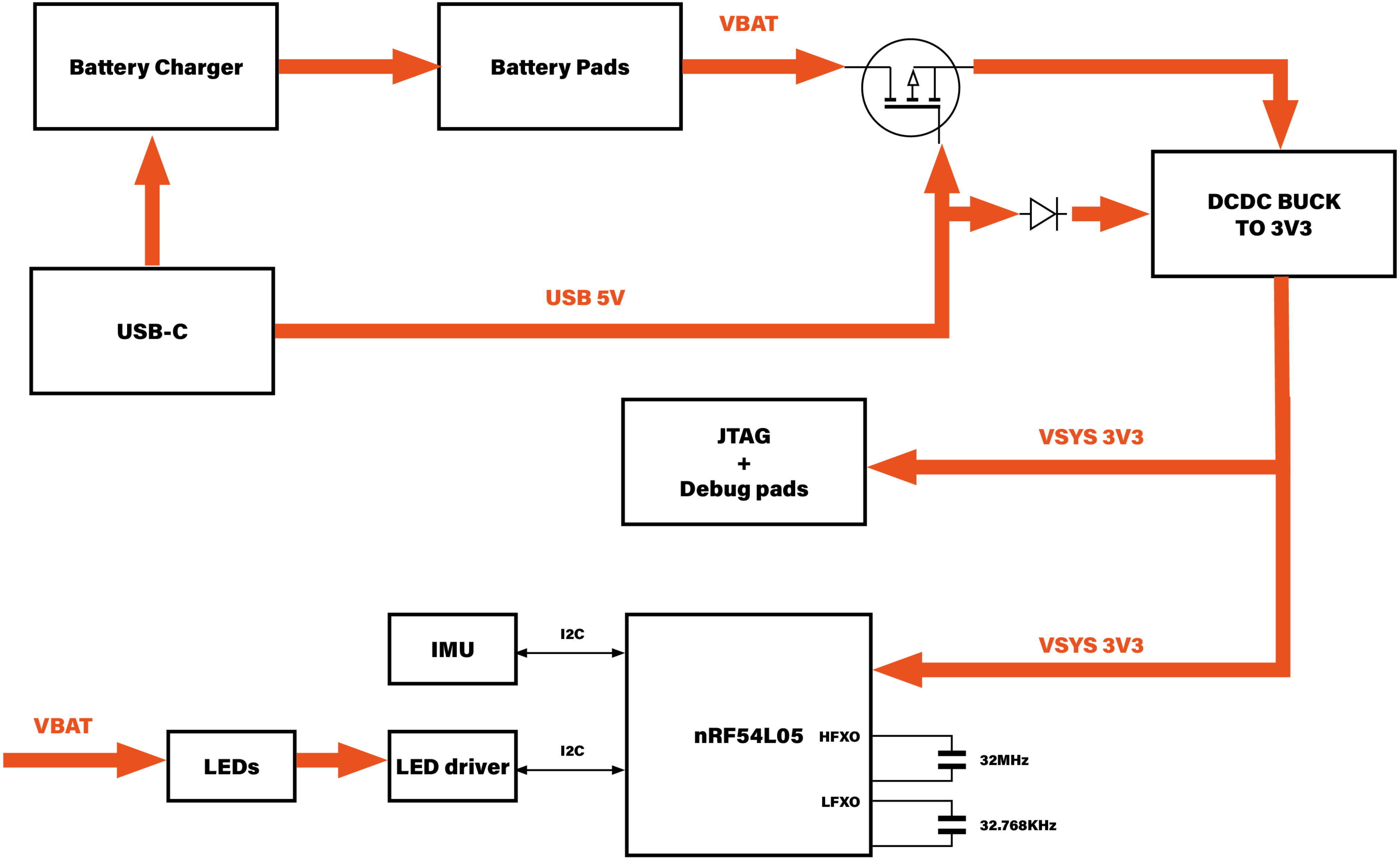
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 1/7

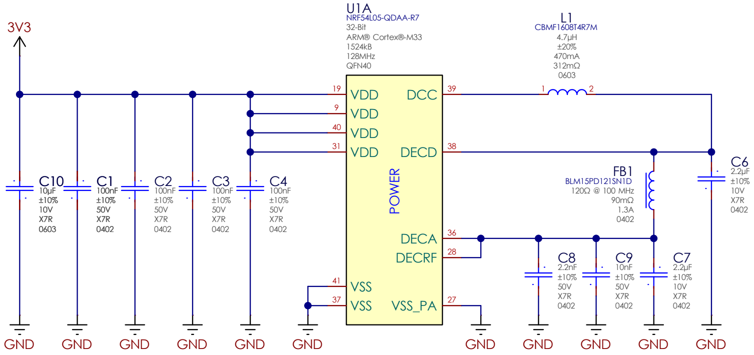
Sheet: /Info/
File: Info.kicad_sch

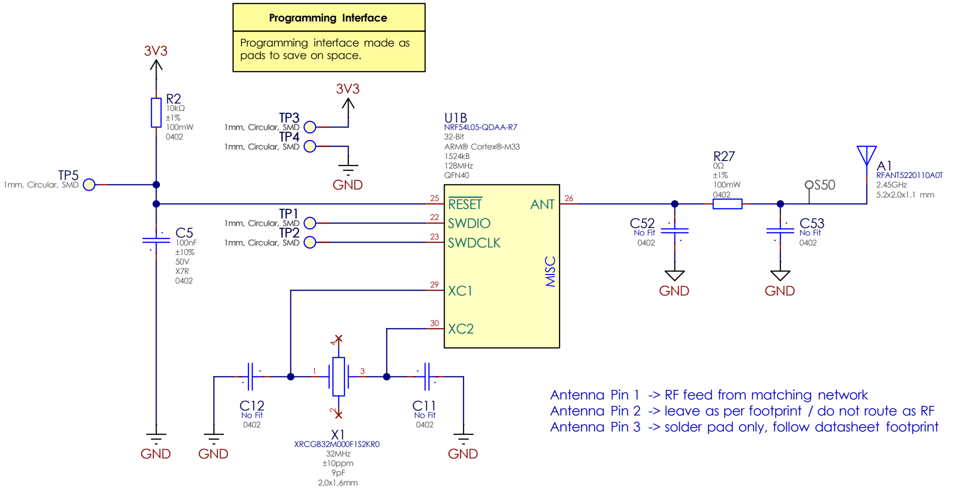
Title:

Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 2/7


Battery Charger
Battery Pads
VBAT
DCDC BUCK
TO 3V3
USB-C
USB 5V
JTAG
+
Debug pads
VSYS 3V3
IMU
I2C
nRF54L05
HFXO
32MHz
LFXO
32.768KHz
VSYS 3V3
VBAT
LEDs
LED driver
I2C

nRF54L05-QDXX

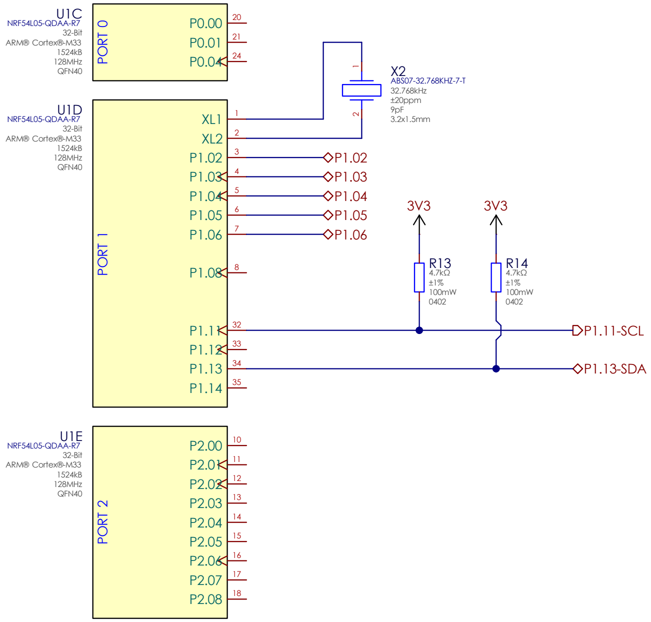

3V3
U1A
NRF54L05-QDAA-R7
32-Bit
ARM® Cortex®-M33
1524kB
128MHz
QFN40
L1
CBMF1608T4R7M
4.7µH
±20%
470mA
312mΩ
0603
VDD 19
VDD 9
VDD 40
VDD 31
DCC 39
DECD 38
DECA 36
DECRF 28
VSS 41
VSS 37
VSS_PA 27
POWER
C10 10µF ±10% 10V X7R 0603
C1 100nF ±10% 50V X7R 0402
C2 100nF ±10% 50V X7R 0402
C3 100nF ±10% 50V X7R 0402
C4 100nF ±10% 50V X7R 0402
FB1
BLM15PD121SN1D
120Ω @ 100 MHz
90mΩ
1.3A
0402
C6 2.2µF ±10% 10V X7R 0402
C8 2.2nF ±10% 50V X7R 0402
C9 10nF ±10% 50V X7R 0402
C7 2.2µF ±10% 10V X7R 0402
GND


3V3
R2 10kΩ ±1% 100mW 0402
TP5 1mm, Circular, SMD
C5 100nF ±10% 50V X7R 0402
TP3 1mm, Circular, SMD
TP4 1mm, Circular, SMD
TP1 1mm, Circular, SMD
TP2 1mm, Circular, SMD
U1B
NRF54L05-QDAA-R7
32-Bit
ARM® Cortex®-M33
1524kB
128MHz
QFN40
25 RESET
22 SWDIO
23 SWDCLK
29 XC1
30 XC2
ANT 26
MISC
R27 0Ω ±1% 100mW 0402
S50
A1
RFANT5220110A0T
2.45GHz
5.2x2.0x1.1 mm
C52 No Fit 0402
C53 No Fit 0402
C12 No Fit 0402
C11 No Fit 0402
X1
XRCGB32M000F1S2KR0
32MHz
±10ppm
9pF
2.0x1.6mm
GND

Programming Interface

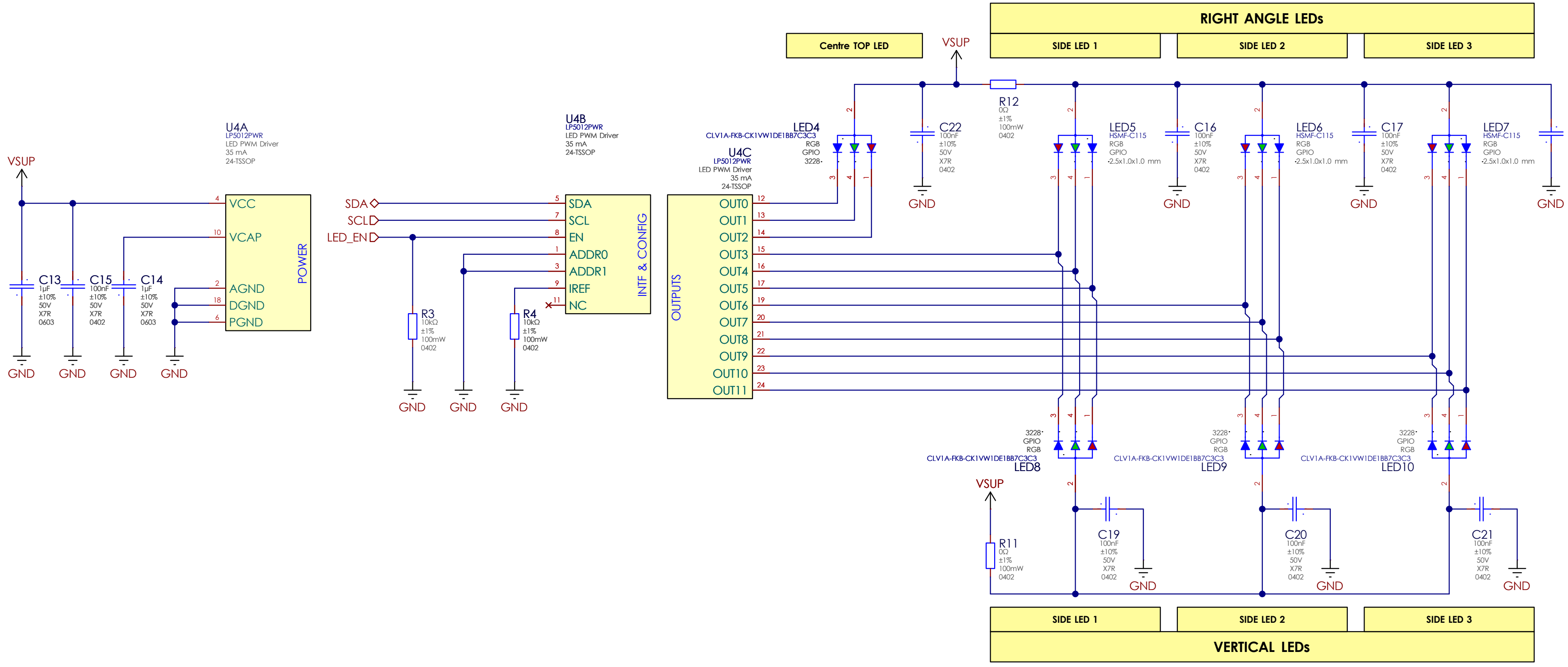
Programming interface made as pads to save on space.

Antenna Pin 1 -> RF feed from matching network
Antenna Pin 2 -> leave as per footprint / do not route as RF
Antenna Pin 3 -> solder pad only, follow datasheet footprint


U1C
NRF54L05-QDAA-R7
32-Bit
ARM® Cortex®-M33
1524kB
128MHz
QFN40
PORT 0
P0.00 20
P0.01 21
P0.04 24
U1D
NRF54L05-QDAA-R7
PORT 1
XL1 1
XL2 2
P1.02 3
P1.03 4
P1.04 5
P1.05 6
P1.06 7
P1.08 8
P1.11 32
P1.12 33
P1.13 34
P1.14 35
X2
ABS07-32.768KHZ-7-T
32.768kHz
±20ppm
9pF
3.2x1.5mm
P1.02
P1.03
P1.04
P1.05
P1.06
3V3
R13 4.7kΩ ±1% 100mW 0402
R14 4.7kΩ ±1% 100mW 0402
P1.11-SCL
P1.13-SDA
U1E
NRF54L05-QDAA-R7
PORT 2
P2.00 10
P2.01 11
P2.02 12
P2.03 13
P2.04 14
P2.05 15
P2.06 16
P2.07 17
P2.08 18

nRF54L15 ANT pin
|
| short 50 Ω RF trace
|
+-- Cshunt_1 to GND, DNP initially
|
Lseries / 0Ω link
|
+-- Cshunt_2 to GND, DNP initially
|
| short 50 Ω RF trace
|
Walsin RFANT5220110A0T Pin 1 / Feed

AETHERFIELD LTD.
Sheet: /nRF54L05-QDXX/
File: mainDesign.kicad_sch
Title:
Size: A4 Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 4/7


VSUP
U4A
LP5012PWR
LED PWM Driver
35 mA
24-TSSOP
VCC
VCAP
AGND
DGND
PGND
POWER
C13
1µF
±10%
50V
X7R
0603
C15
100nF
±10%
50V
X7R
0402
C14
1µF
±10%
50V
X7R
0603
GND
U4B
LP5012PWR
LED PWM Driver
35 mA
24-TSSOP
SDA
SCL
LED_EN
EN
ADDR0
ADDR1
IREF
NC
INTF & CONFIG
R3
10kΩ
±1%
100mW
0402
R4
10kΩ
±1%
100mW
0402
U4C
LP5012PWR
LED PWM Driver
35 mA
24-TSSOP
OUTPUTS
OUT0
OUT1
OUT2
OUT3
OUT4
OUT5
OUT6
OUT7
OUT8
OUT9
OUT10
OUT11
Centre TOP LED
RIGHT ANGLE LEDs
SIDE LED 1
SIDE LED 2
SIDE LED 3
LED4
CLV1A-FKB-CK1VW1DE1BB7C3C3
RGB
GPIO
3228
C22
100nF
±10%
50V
X7R
0402
R12
0Ω
±1%
100mW
0402
LED5
HSMF-C115
RGB
GPIO
2.5x1.0x1.0 mm
C16
100nF
±10%
50V
X7R
0402
LED6
HSMF-C115
RGB
GPIO
2.5x1.0x1.0 mm
C17
100nF
±10%
50V
X7R
0402
LED7
HSMF-C115
RGB
GPIO
2.5x1.0x1.0 mm
C18
100nF
±10%
50V
X7R
0402
LED8
LED9
LED10
R11
0Ω
±1%
100mW
0402
C19
C20
C21
VERTICAL LEDs

Sheet: /LED/
File: LED.kicad_sch

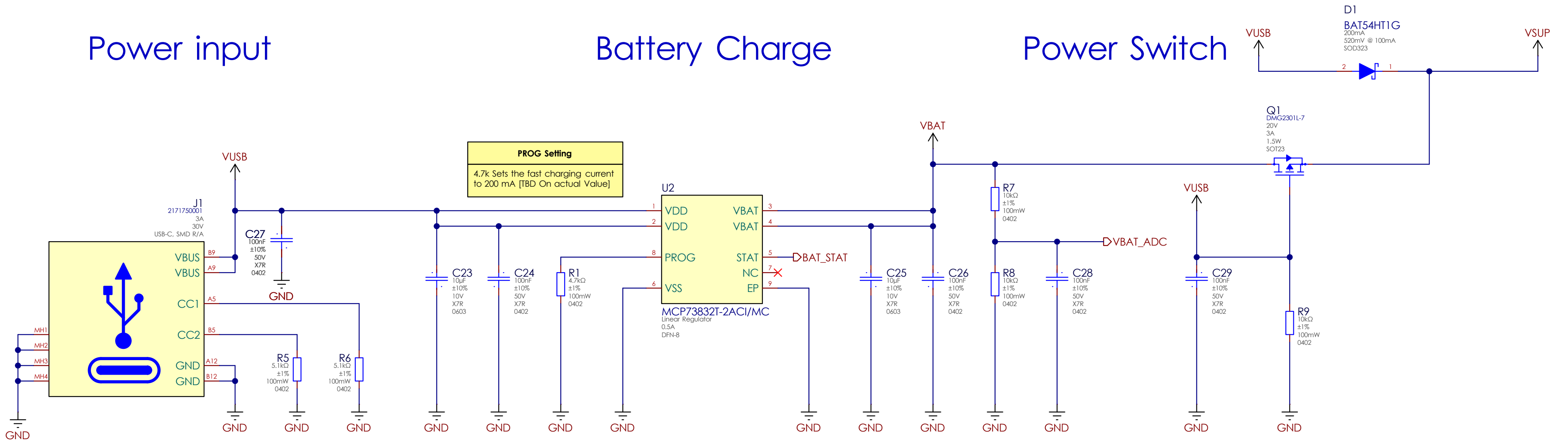
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 5/7

Power input

Battery Charge

Power Switch


J1
2171750001
3A
30V
USB-C, SMD R/A
VBUS
VBUS
CC1
CC2
GND
GND
B9
A9
A5
B5
A12
B12
MH1
MH2
MH3
MH4
VUSB
C27
100nF
±10%
50V
X7R
0402
R5
5.1kΩ
±1%
100mW
0402
R6
5.1kΩ
±1%
100mW
0402
C23
10µF
±10%
10V
X7R
0603
C24
100nF
±10%
50V
X7R
0402
R1
4.7kΩ
±1%
100mW
0402
U2
VDD
VDD
PROG
VSS
VBAT
VBAT
STAT
NC
EP
1
2
8
6
3
4
5
7
9
MCP73832T-2ACI/MC
Linear Regulator
0.5A
DFN-8
BAT_STAT
VBAT
C25
10µF
±10%
10V
X7R
0603
C26
100nF
±10%
50V
X7R
0402
R7
10kΩ
±1%
100mW
0402
R8
10kΩ
±1%
100mW
0402
C28
100nF
±10%
50V
X7R
0402
VBAT_ADC
VUSB
C29
100nF
±10%
50V
X7R
0402
R9
10kΩ
±1%
100mW
0402
Q1
DMG2301L-7
20V
3A
1.5W
SOT23
D1
BAT54HT1G
200mA
520mV @ 100mA
SOD323
VUSB
VSUP
2
1
GND

PROG Setting

4.7k Sets the fast charging current to 200 mA [TBD On actual Value]

System 3.3V

1V8 Buck Module

Input Capacitance

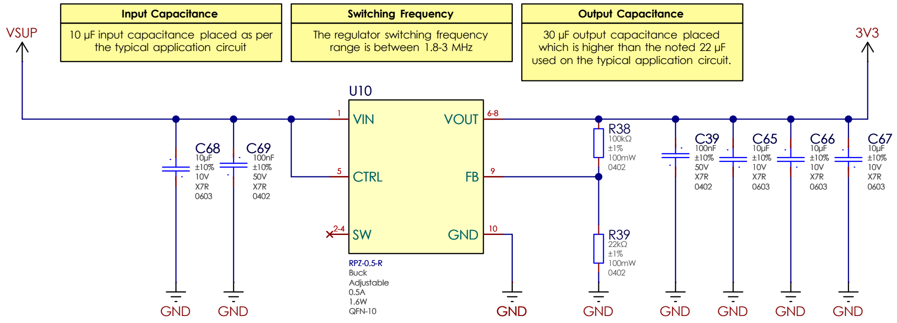
10 µF input capacitance placed as per the typical application circuit

Switching Frequency

The regulator switching frequency range is between 1.8-3 MHz

Output Capacitance

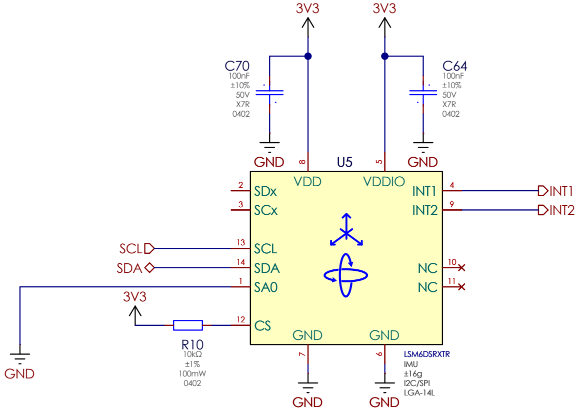
30 µF output capacitance placed which is higher than the noted 22 µF used on the typical application circuit.


VSUP
C68
10µF
±10%
10V
X7R
0603
C69
100nF
±10%
50V
X7R
0402
U10
VIN
CTRL
SW
VOUT
FB
GND
1
5
2-4
6-8
9
10
RPZ-0.5-R
Buck
Adjustable
0.5A
1.6W
QFN-10
R38
100kΩ
±1%
100mW
0402
R39
22kΩ
±1%
100mW
0402
C39
100nF
±10%
50V
X7R
0402
C65
10µF
±10%
10V
X7R
0603
C66
10µF
±10%
10V
X7R
0603
C67
10µF
±10%
10V
X7R
0603
3V3
GND

Sheet: /Power/
File: Power.kicad_sch

Title:

Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 6/7


3V3
3V3
C70
100nF
±10%
50V
X7R
0402
C64
100nF
±10%
50V
X7R
0402
GND
GND
U5
8
5
VDD
VDDIO
2
SDx
3
SCx
4
INT1
9
INT2
INT1
INT2
SCL
13
SCL
SDA
14
SDA
1
SA0
NC
10
NC
11
3V3
12
CS
GND
GND
R10
10kΩ
±1%
100mW
0402
7
6
LSM6DSRXTR
IMU
±16g
I2C/SPI
LGA-14L
GND
GND
GND

```
      IMU_3V3
         |
    +-----+------+
    |            |
   VDD         VDDIO
    |            |
    |      LIS2DW12TR
    |            |
   GND---------GND

nRF P0.03 / IMU_SCL  ---- SCL/SPC
nRF P0.04 / IMU_SDA  ---- SDA/SDI

IMU_3V3 -- 4.7k -- SCL
IMU_3V3 -- 4.7k -- SDA

CS --------------------  IMU_3V3
SDO/SA0 ----------------  GND

INT1 -------------------  nRF P0.02 / IMU_INT1
INT2 -------------------  NC or spare GPIO
```

SA0

Pulling SA0 low sets the device I2C 7 bit address to 0x68. See section 7.2.4.2 of the datasheet.

Sheet: /Sensors/
File: Sensors.kicad_sch

Title:

Size: A4 Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 7/7